



Robust Adversarial Defence: Use of Auto-inpainting

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Abstract. Adversarial patch attacks have become a primary concern in recent years as they pose a significant threat to the security and reliability of deep neural networks. Modifying benign images by introducing adversarial patches comprising localized adversarial pixels alters the salient features of the image resulting in misclassification. The novelty of our approach is in the use of image inpainting technique as an adversarial defence for rectifying the patch region. Adversarial patch is automatically localized using Fast Score Class Activation Map and superseded by inpainting using Fast Marching Method which efficiently propagates pixel information from the surrounding areas into the patch region. This approach ensures original image's structural integrity while simultaneously inpainting the adversarial pixels. Moreover, at the time of the attack it is not expected to have prior knowledge about the patch. Therefore, we propose our novel adversarial defence technique in a black-box setting assuming no knowledge about the patch location, shape or its size. Furthermore, we do not rely on re-training our victim model on adversarial examples, indicating its potential usefulness for real-world applications. Our experimental results show that the proposed approach achieves accuracy up to 76.37% on ImageNet100 despite the adversarial patch attack amounting to a considerable improvement of 76.28% points. Moreover, on benign images our approach gives decent accuracy of 81.11% thereby suggesting that our defence pipeline is applicable irrespective of whether the input image is adversarial or clean.

Keywords: Adversarial Machine Learning · Adversarial Defence · Inpainting

1 Introduction

The increasing use of deep neural networks has led to a growing concern regarding adversarial attacks which involve adding small perturbations to the input

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data that may be imperceptible to humans but can significantly alter the prediction of the machine-learning models as shown in Eq. (1).

$$f(g(x, \alpha)) = t \text{ subject to } \|x - g(x, \alpha)\|_p \leq \epsilon, \quad (1)$$

where x is the input image, f is the classifier, t is the target class for misclassification, g is the perturbation function applied on x , α is a parameter and ϵ is a threshold.

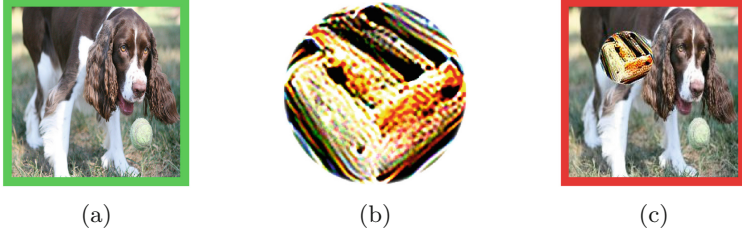


Fig. 1. Patch-based attack. (a) Benign image correctly classified as *dog* (b) Adversarial patch obtained by setting the target class as *toaster*. (c) Adversarial image generated by superimposing scaled and rotated (b) on (a) gets misclassified into *toaster*. Red and Green boxes signify incorrect and correct predictions, respectively. (Color figure online)

More recently, patch-based adversarial attacks [3, 5, 7] have gained the attention of researchers which occupy only a small region in an image and may not raise any suspicion amongst humans, however, they pose an equally significant threat to machine learning models, Fig. 1 illustrates the *adversarial patch* attack [1]. Such attacks can also be realized physically by pasting a printed patch on the target object to be misclassified. Hence, it is crucial to develop an adversarial defence against them. To the best of our knowledge, the defences proposed to counter such attacks are in white-box settings which assume complete knowledge of the patch such as patch size, location and attack type. However, black-box defence is practical as it allows defenders to respond quickly to new threats without needing complete knowledge of the patch. Hence, we propose a novel adversarial defence in the black-box setting.

Determining whether the image is benign or adversarial is the foremost step in defending against an adversarial patch. For this, several patch detection methods have been successful in the literature [16, 17, 19]. However, little attention has been given to improving the model’s classification accuracy in spite of the adversarial patch attack. Thus, our defence mechanism is designed first to localize the adversarial patch automatically using Fast Score Class Activation Mapping (FSCAM) [9], an advanced version of Score Class Activation Mapping (SCAM) [15] that generates attention maps highlighting the region in an image that may significantly influence classification decision in turn, localizing the patch. Thus, enabling accurate localization and defence against potential attacks.

Further, we utilize a well-established in-painting technique known as Fast Marching Method (FMM) [14] often used by researchers working in the area of image restoration for repairing damaged parts of an image [13]. The method involves inpainting an image’s missing regions using information from the surrounding areas. To the best of our knowledge, researchers have not attempted the use of inpainting techniques for defending against adversarial attacks. Our contributions can be summarized in two main areas:

1. We present a novel adversarial defence technique: *Auto-Inpainting* based on image inpainting technique.
2. Our defence is devised in black-box setting and does not necessitate re-training the victim model.

2 Related Work

Patch attacks are a significant concern, enabling attackers to alter images by arbitrarily modifying their pixels. One such threatening attack is LaVAN [5] which creates localized visible adversarial noise causing classifiers to misclassify images to arbitrary labels in the digital domain. However, there are potential limitations to LaVAN when transferring the noise which may affect its practicality in real-world scenarios. In contrast, the adversarial patch by Brown et al. [1] is another patch-based attack that creates robust and targeted adversarial image patches in the physical world using attachable stickers to cause classifiers to output any target class. These stickers can cause significant harm to vision systems such as object detection [4] and visual tracking [3]. Hence, in our work, we consider patch-based attack proposed by Brown et al. [1].

Patch detectors are essential for detecting adversarial patches and have received much attention recently [10, 11, 18, 20]. Patch detectors are trained on a dataset of images with adversarial patches, and their accuracy in finding the patch location is crucial for the security of a machine-learning model. After the patch is located, creating certified defences that protect against all possible attacks for a given threat model is essential. For instance, Chiang et al. [2] developed an adversarial defence against patch attacks using Interval Bound Propagation (IBP) on MNIST and CIFAR10, but it is challenging to scale it to ImageNet. Another approach is Derandomized Smoothing (DS) [8], which has improved accuracy compared to IBP on ImageNet but requires expensive computational resources. Majority of the techniques [16, 17] operate under white-box defence setting requiring prior knowledge of the attack patch size. In comparison, our approach can defend against robust patch-based attacks of any patch shape and size without prior knowledge with less computation and without retraining the model.

3 Proposed Auto-inpainting Method

Figure 2 shows the overview of the Auto-Inpainting approach. Given an image x_{in} , we generate a heatmap ($hmap$) of salient features with the help of FSCAM,

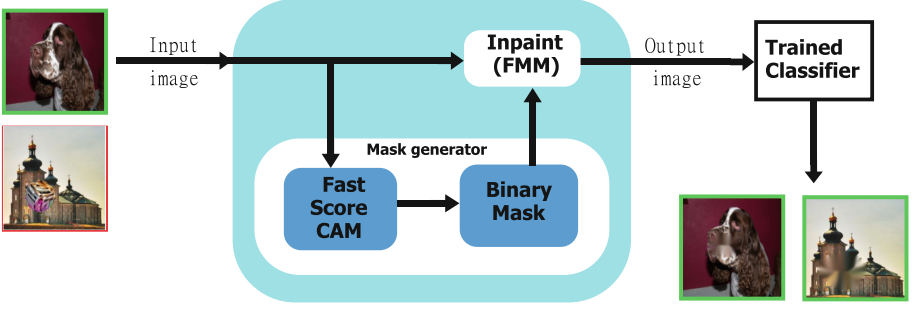


Fig. 2. Overview of the proposed Auto-Inpainting method.

which is then converted into a binary mask (m) with a pre-defined threshold (ϵ_p). Further, the image x_{in} along with the binary mask m are given as input to our inpainting function. Finally, the inpainted image is fed to the trained classifier. Note that our approach works in the black box setting, assuming no knowledge about whether the image is benign or adversarial, the shape and size of the patch or its location. Furthermore, we do not modify the original classifier trained on benign images in any way as opposed to adversarial training.

3.1 Adversarial Patch Generation

Suppose $x \in \mathcal{R}^{H \times W \times C}$ to be a clean image correctly classified by the model $f(x) = y$, where y is the correct class label and H , W and C represent the image height, width and number of channels, respectively. The adversary aims to push x to a specific class C_t under a targeted attack or any other class C_i in an untargeted attack setting. As described by Brown et al. [1], these attacks can be realized using a patch application operator, denoted as $\zeta(p, x, l, t)$, for a given image x , patch p , patch location l , and patch transformations t . The operator applies the transformations t to the patch p and then applies the transformed patch p to the image x at location l to generate perturbed image $x' \in \mathcal{R}^{H \times W \times C}$.

3.2 Adversarial Defence Using Auto-Inpainting

FSCAM is a promising technique that has not been utilized for patch localization yet. It outperforms other methods like Grad-Cam and Grad-Cam++ [9] because it does not rely on gradients and has a more reasonable weight representation. Using this method, we generate the class activation map (CAM) that highlights the salient regions in the input image primarily responsible for the prediction. Let x_{in} be any input image, benign or adversarial, $f(x_{in}) = y_{in}$, where y_{in} is the predicted class label. In this case, FSCAM is expected to localize the patch region in x_{in} that led to the classification.

Further, a binary mask $m \in \mathcal{R}^{H \times W \times C}$ is generated by thresholding where patch pixels are assigned ones (m_1) and non-patch pixels are assigned zeros (m_0).

Thus, m_1 represents the region that needs to be inpainted in the adversarial image by the FMM algorithm, a simple yet efficient inpainting technique. FMM [14] is a well-established technique to fill in missing or damaged portions of an image with surrounding information, but it has not been explored for adversarial defence. This has motivated us to propose a novel approach that leverages FMM for adversarial defence as it retains the features required for correct classification. Our approach uses the FMM algorithm that first identifies the missing region with the mask m and estimates the missing pixels' values using the nearby patch region in the input image, then uses them in inpainting as in Eq. 3 to get $x_{inpaint}$. For notations used in the Eq. 3, refer Algorithm 1

$$x_{inpaint} = \frac{\sum_{q \in B_{\Delta}(p)} w(p, q)(x_{in}(q) + \nabla x_{in}(q)(x_{in}(p) - x_{in}(q)))}{\sum_{q \in B_{\Delta}(p)} w(p, q)} \quad (2)$$

where

$$w(p, q) = \text{dir}(p, q) \cdot \text{dst}(p, q) \cdot \text{lev}(p, q) \quad (3)$$

where $\text{dir}(p, q)$ ensures a higher contribution of pixels close to the normal direction at p , $\text{dst}(p, q)$ decreases the contribution of pixels geometrically farther from p and $\text{lev}(p, q)$ ensures that pixels close to the contour through p contribute more than farther pixels.

Algorithm 1: Auto-Inpainting Algorithm

Input: input image (x_{in}), binary mask(m) containing m_1 , region to be inpainted

Parameters: threshold (ϵ_p), surrounding region outside m_1 ($B_{\Delta}(p)$)

Output: inpainted image ($x_{inpaint}$)

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1  $\delta m_i = \text{Boundary pixels of region } m_1$ 
2  $\delta m = \delta m_i$ 
3 while ( $\delta m$  not empty) do
4    $q = \text{pixel in } x_{in} \text{ closest to first pixel in } \delta m \text{ and } \in B_{\Delta}(p),$ 
5    $p = \text{pixel in } x_{in} \in \delta m \text{ closest to } q$ 
6    $x_{inpaint}(p) = \text{inpaint}(p, q)$  using the Eq. 3
7   Advance  $\delta m$  into  $m_1$            # Marching into  $m_1$  region
8 end while
9 return  $x_{inpaint}$ 
```

4 Experimental Results

In this section, we outline the necessary experimental setup and define metrics for evaluating our approach in a black-box setting. We then present a comprehensive analysis of our defence pipeline's performance, adjusting various parameters and discussing both successes and failures.

4.1 Experimental Setup

We consider the Imagenet dataset, particularly validation data consisting of 5000 images belonging to 100 classes with 50 images per class. Out of the 5000 images, we randomly sampled 3000 images for our experiment. The images in the dataset are of different sizes, hence we resized them to (224, 224, 3). Also, we generated our patch by training it on a ResNet50 model (not the victim model) and then superimposed it at random locations with a scale factor of 0.3, roughly 7.6% of the image size, to generate our adversarial examples. Further, we used a pre-trained ResNet50 model trained on the Imagenet dataset, a.k.a the victim model, to evaluate the robustness of the proposed *Auto-Inpainting* method. The class activation maps have been generated using FSCAM GitHub implementation [6] followed by binarizing the map with a threshold ϵ_p set to 0.5. We also experiment with different values of ϵ_p to analyze how it affects our model accuracy.

4.2 Experimental Results and Discussion

In the first set of experiments, we evaluate the vulnerability of the pre-trained ResNet50 model to the patch-based attacks. Particularly, we report the attack success rate (ASR) which is the proportion of successful patch-based attacks. As can be seen from Table 1(a), very high value of (ASR) indicates the urgency with which these attacks need to be addressed.

Authors of FSCAM observed poor performance at generating correct heatmaps when softmax is used as an output layer activation function (AF) [12]. Therefore, While generating activation maps, we have replaced the softmax in the output layer with three widely-used AFs namely, linear, relu and tanh. Table 1(a) shows the results. A small increase can be noticed in the Auto-Inpainting accuracy when we used tanh as AF at the output layer. Note that these AFs are used only while generating heatmaps while at the time of classification softmax AF is retained.

Table 1. Auto-Inpainting accuracy

Output layer AF	ASR	ϵ_p	Adversarial Accuracy	Benign Accuracy	ϵ_p	Adversarial Accuracy
linear	99.91%	0.5	71.57%	67.74%	0.5	71.54%
relu	99.91%	0.5	71.57%	67.84%	0.6	74.74%
tanh	99.91%	0.5	72.39%	65.11%	0.7	76.34%
tanh	99.91%	0.7	76.37%	81.11%	0.75	74.68%

(a) Different AFs at FSCAM output layer

(b) Different values of ϵ_p for linear AF

In the second set of experiments, we evaluated the robustness of our Auto-Inpainting method for patch-based attacks. Table 1(a) shows the *Adversarial Accuracy and Benign Accuracy* computed as the proportion of the no. of correctly classified inpainted images among the total number of tested adversarial images and benign images, respectively. Black-box adversarial defences are generally harder to devise than white-box defences owing to no prior knowledge about the patch size, shape, or location in the original image. However, in the black-box setting, by fine-tuning the parameters, we are able to achieve an accuracy of 76.34%, refer to Table 1(b). Figure 3 helps visualize how the image looks at each stage of our pipeline. Moreover, as can be observed from Fig. 3(c), we were able to detect the salient features of any shape, which makes our approach robust to patches of different shapes and size as well.

Despite extensive research on white-box defence techniques, there has been limited progress in developing effective black-box defences. As far as we know, researchers have not attempted defence techniques with the kind of settings we have considered. Authors in PatchZero [19] proposed an adversarial defence in the white-box setting and considered very small patch size, a rectangular box with 2% pixels of the image size, and achieved the highest accuracy of 76.80% on the same dataset. On the other hand, the patch size considered in our work is $\approx 7.6\%$ of the image size. Moreover, their defence re-trains the victim model, while our approach does not. This suggests that our defence pipeline is effective in real-world scenarios where we generally don't have any prior information about the patch. It is important to note that there is still much work to be done in this field, but our findings may be a step in the right direction.

Additionally, since our approach is expected to work in a black-box defence setting, it should work for benign images in addition to adversarial images. Therefore, we have also computed the classification accuracy on benign images in the presence of the Auto-Inpainting defence. Although there is a decrease in benign accuracy, on fine-tuning the parameters, we could attain an accuracy of 81.11%, refer to Table 1(a). It is worth mentioning that while we are able to correctly classify the image, a decrease in confidence score was observed.

Effective localization of the patch in the adversarial image is crucial to the efficacy of our adversarial defence technique. Therefore, in order to evaluate the effectiveness of patch localization by FSCAM, we computed *Intersection over Union* (IoU) between the ground truth and the predicted binary mask. IoU can be defined as the ratio of the area of the overlapping region to the area of the union of the two masks. We obtained an *IoU* score of 0.483 at the threshold $\epsilon_p = 0.5$. As can be observed from Fig. 4 the low IoU suggests that the generated mask covers the entire patch region as well as the outer region consisting of benign pixels. This may adversely affect the accuracy of the model. Therefore, we considered different values of ϵ_p from 0.5 to 0.7 and observed considerable improvement in the accuracy from 71.5% to 76.34% as shown in Table 1(b). Also, as can be seen from the figure, on increasing ϵ_p , the size of the mask region that we generate shrinks, reducing the area to inpaint in turn resulting in less information loss during inpainting process.

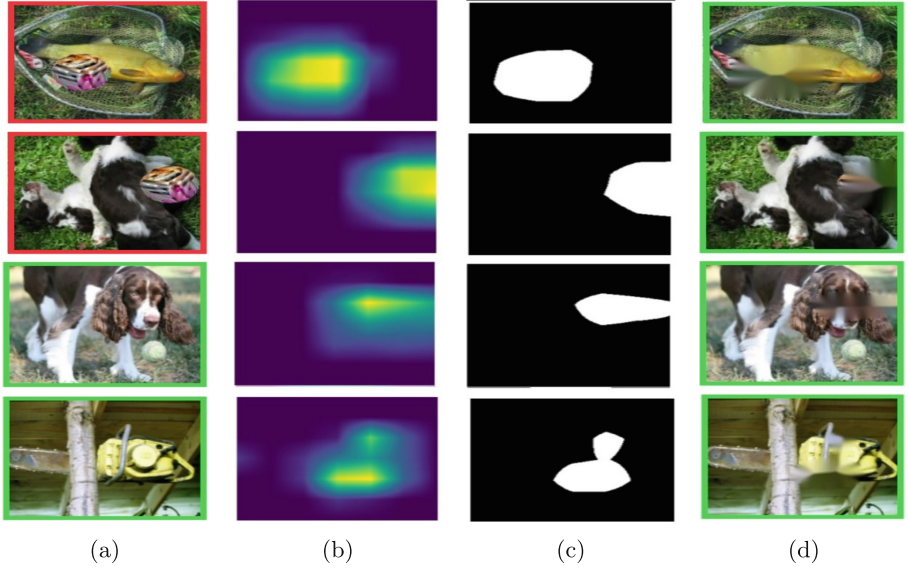


Fig. 3. Qualitative results: (Green and Red boxes represent correctly and incorrectly classified inputs to our victim model, respectively) First two and last two rows show the results for adversarial examples and benign images, respectively. **Note that the model predicts the class of adversarial images correctly after *Auto-Inpainting*.** (Color figure online)

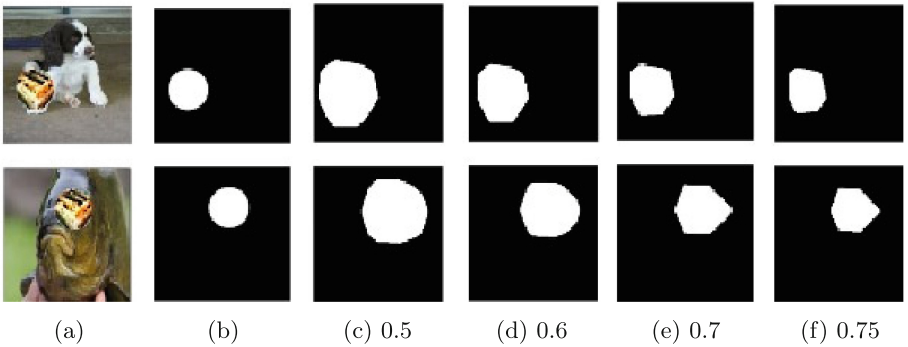


Fig. 4. Figure illustrating effect of ϵ_p value on the binary mask. (a) Adversarial examples. (b) Ground truth binary mask. (c) to (f) binary masks corresponding to different values of ϵ_p .

Failure Cases and Limitations. Finally, we present two failure cases of our method in Fig. 5. First, if the patch appears over the salient regions in the scene, it becomes even more difficult to restore those pixel values. For instance, in Fig. 5(a), the patch has been placed on the major part of the object in the image and the inpainted image loses much of the information resulting in incorrect

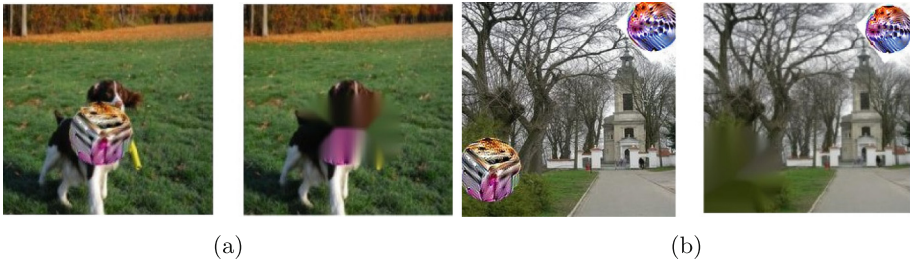


Fig. 5. (a) The adversarial patch covers the main object in the scene making it difficult to get correct prediction after inpainting. (b) The defence pipeline fails to detect more than two different patches at once.

classification. The second case is when we apply two different patches, only one of the patches is localized by the FSCAM method. For instance, in Fig. 5(b) two patches generated from target classes *toaster* and *cannon* have been placed on the benign image, however, only **toaster** patch was localized and inpainted by our approach leading the **cannon** patch to misclassify the adversarial image.

5 Conclusion and Future Work

In this paper, we proposed a novel adversarial defence technique *Auto-Inpainting* motivated by use of image inpainting techniques for image restoration tasks. The defence is carried out in a black-box setting without assuming the knowledge of the patch size, shape as well as location. Moreover, the proposed approach works well for both adversarial as well as benign inputs. *Auto-Inpainting* detects the adversarial region using the Fast-Score CAM technique to generate class activation map and then binarize the heatmap to localize the patch region. Further, the Fast Marching Method technique used for inpainting restores the patch pixels using the surrounding pixels. Our approach achieves an accuracy of 76.37% and 81.11% for adversarial and benign cases, respectively. Our future work involves considering different CNN models and attack variants to test the efficacy of our approach. Moreover, more sophisticated patch localization and inpainting techniques can be explored to make the approach even more robust.

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